NLP Assignment 1: Solutions

PAN CHANGXUN
Student ID: 2024011323

August 2026

Problem 1: CFG

1. Chomsky Normal Form

There are no ϵ-productions. Eliminate the unit productions

$$\text{NP} \to \text{PropN}, \quad \text{Nom} \to \text{Noun}, \quad \text{VP} \to \text{Verb}$$

by copying their terminal rules, then remove the unreachable PropN. An equivalent CNF grammar is

$$\begin{aligned}
S &\to \text{NP VP}, \\
\text{NP} &\to \text{Det Nom} \mid \texttt{Houston}, \\
\text{Nom} &\to \text{Noun Nom} \mid \texttt{book} \mid \texttt{flight}, \\
\text{VP} &\to \text{Verb NP} \mid \texttt{booked}, \\
\text{Det} &\to \texttt{the} \mid \texttt{a}, \\
\text{Noun} &\to \texttt{book} \mid \texttt{flight}, \\
\text{Verb} &\to \texttt{booked}.
\end{aligned}$$

Every rule is now of the form $A \to BC$ or $A \to a$.

2. CKY for "the book booked"

For $w_1 = \texttt{the}$, $w_2 = \texttt{book}$, and $w_3 = \texttt{booked}$, the completed chart is

	$j=1$	$j=2$	$j=3$
$i=1$	$\{\text{Det}\}$	$\{\text{NP}\}$	$\{S\}$
$i=2$		$\{\text{Noun}, \text{Nom}\}$	$\varnothing$
$i=3$			$\{\text{Verb}, \text{VP}\}$

$C_{1,2}$ contains NP because NP $\to$ Det Nom, and $C_{1,3}$ contains S because $S \to$ NP VP. Thus $S \in C_{1,3}$ and the sentence is **accepted**.

Problem 2: Bigram Model

1. Add-one smoothing

The vocabulary is

$$\{\texttt{<s>}, \texttt{I}, \texttt{am}, \texttt{Sam}, \texttt{</s>}, \texttt{do}, \texttt{not}, \texttt{like}, \texttt{green}, \texttt{eggs}, \texttt{and}\},$$

so $|V| = 11$. Since $C(\texttt{am}) = 3$ and $C(\texttt{am Sam}) = 2$,

$$P(\texttt{Sam} \mid \texttt{am}) = \frac{2+1}{3+11} = \boxed{\frac{3}{14} \approx 0.2143}.$$

2. Model without an end symbol

The unsmoothed estimates are

$$P(\texttt{a} \mid \texttt{<s>}) = P(\texttt{b} \mid \texttt{<s>}) = P(\texttt{a} \mid \texttt{a}) = P(\texttt{b} \mid \texttt{a}) = P(\texttt{a} \mid \texttt{b}) = P(\texttt{b} \mid \texttt{b}) = \frac{1}{2}.$$

Each two-word string has probability $(1/2)^2 = 1/4$, so

$$\sum_{w_1, w_2 \in \{\texttt{a},\texttt{b}\}} P(w_1 w_2) = 4 \cdot \frac{1}{4} = \boxed{1}.$$

Each three-word string has probability $(1/2)^3 = 1/8$, so

$$\sum_{w_1, w_2, w_3 \in \{\texttt{a},\texttt{b}\}} P(w_1 w_2 w_3) = 8 \cdot \frac{1}{8} = \boxed{1}.$$

Thus every fixed length receives total mass 1; over all lengths the mass is $\sum_{n \geq 1} 1 = \infty$. Without a stopping symbol, this is not a probability distribution over finite sentences.

Problem 3: RLHF

1. Optimal policy

Fix x, and let $q(y) = \pi_{\text{ref}}(y \mid x)$ and $r_y = r(x, y)$. Define

$$Z(x) = \sum_{y'} q(y')e^{r_{y'}/\beta}, \qquad \pi^*(y \mid x) = \frac{q(y)e^{r_y/\beta}}{Z(x)}.$$

For any π,

$$\mathrm{KL}(\pi\|\pi^*) = \mathrm{KL}(\pi\|q) - \frac{1}{\beta}\sum_{y} \pi(y \mid x)r_y + \log Z(x),$$

so the per-prompt objective satisfies

$$\sum_{y} \pi(y \mid x)r_y - \beta\,\mathrm{KL}(\pi\|q) = \beta \log Z(x) - \beta\,\mathrm{KL}(\pi\|\pi^*).$$

Since KL divergence is nonnegative, the maximum is attained at $\pi = \pi^*$. The unrestricted full objective separates over prompts, so this holds for every x, proving Eq. (3)–(4).

2. Direct calculation

No. Computing $Z(x)$ requires summing over all possible response sequences and evaluating their rewards, which is computationally intractable. *(20 words)*

3. Deriving Eq. (7)

Rearranging Eq. (3) gives

$$r(x, y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x).$$

Set $\pi^* = \pi_\theta$. For responses paired under the same prompt, the $\beta \log Z(x)$ terms cancel:

$$r(x, y_w) - r(x, y_l) = \beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}.$$

Substitution into Eq. (6) yields

$$\mathcal{L}_\pi(\theta) = -\frac{1}{N}\sum_{i=1}^{N} \log \sigma\left(\beta \log \frac{\pi_\theta(y_w^{(i)} \mid x^{(i)})}{\pi_{\text{ref}}(y_w^{(i)} \mid x^{(i)})} - \beta \log \frac{\pi_\theta(y_l^{(i)} \mid x^{(i)})}{\pi_{\text{ref}}(y_l^{(i)} \mid x^{(i)})}\right),$$

which is Eq. (7).

Problem 4B: Code Completion

I completed all four TODOs in `problem_word2vec/word2vec.py`: the two embedding layers, the negative-sampling loss, frequency-based negative sampling, and the training loop. The completed source file is submitted separately as required.

Problem 4C: Unity 6 Tutorial

1. What I learned

The most interesting part of the tutorial was seeing how Unity combines transforms, colliders, rigid-body physics, and materials. I built ramps and stairs from simple cubes, and the robot interacted with them through gravity and collision instead of passing through the geometry. I also learned that materials tile across resized objects, which makes rapid prototyping easier. The tutorial further showed how lighting, post-processing, VFX, collectibles, UI, and animation can turn a basic blockout into a playable scene.

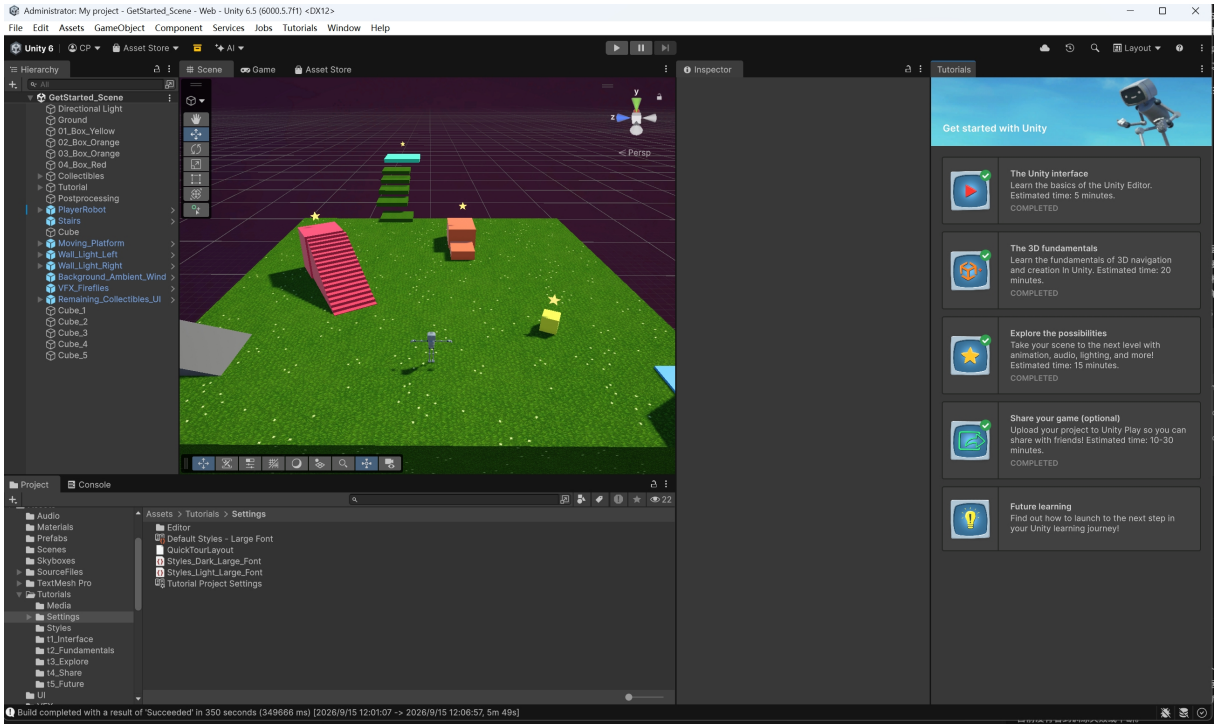
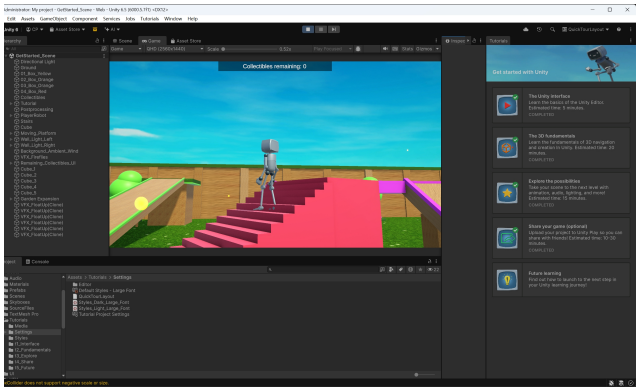


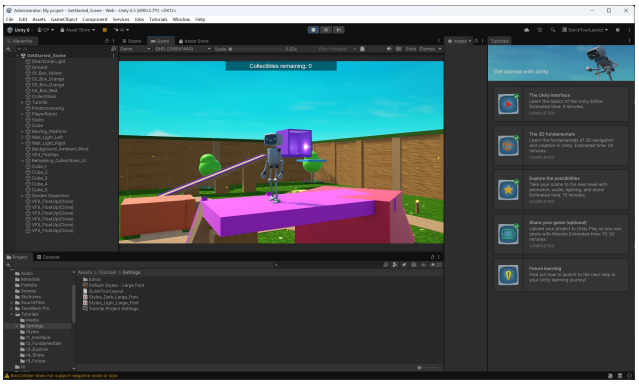
Figure 1: The final robot scene after completing the required tutorial sections.

2. Scene extension

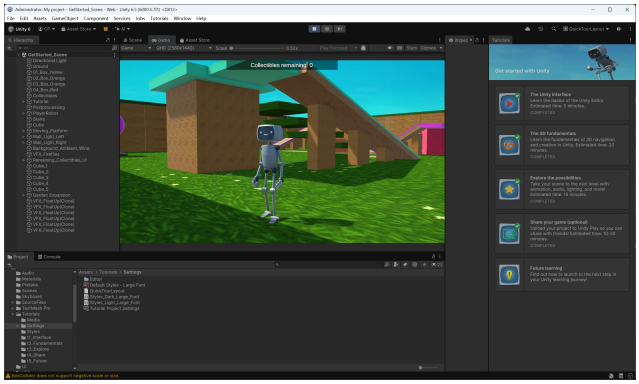
I expanded the play area to 66×74 units and surrounded it with grass-topped collision walls so the robot stays inside the scene. I added more routes to the collectibles, including ramps, stairs, and elevated platforms. I also laid out paths, planted trees and flower beds, and adjusted the lighting, sky, and visual effects to make the level feel more like a small park.



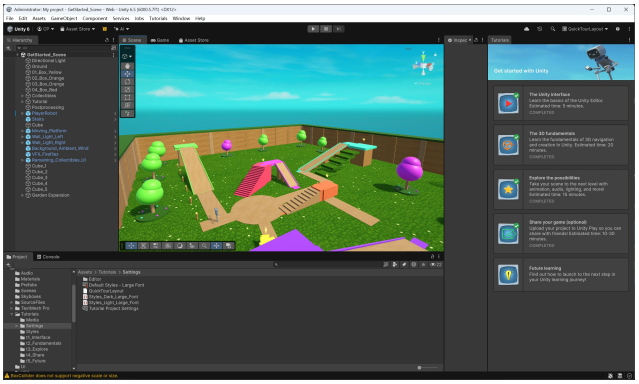
(a) Stair route and enclosed play area



(b) Elevated platform and ramp



(c) Supporting structures below the ramps



(d) Overview of the expanded garden scene

Figure 2: Four views of the extended robot scene.